

Technical Data Sheet

Date of issuing : 01.07.2025
Product Name : Transformer Oil (Naphthenic) - TRANSOL GIS I
Grade : IS : 335 : 2018 – Type – I (I) Uninhibited

Rev. No. 2

Sr. No.	Property	Test Method	Limits	Typical values
A - Function :				
i)	Viscosity at 40 °C	IS :1448 (Part 25)	12 mm ² /s, Max	10.5
ii)	Viscosity at 0 °C	IS :1448 (Part 25)	-	NA
iii)	Viscosity at - 30 °C	IS : 1448 (Part 25)	1800 mm ² /s, Max	1050
iv)	Viscosity at - 40 °C	IS : 16084	-	NA
v)	Pour Point	IS : 1448 (Part 10/Sec2)	- 40 °C, Max	- 42
vi)	Water Content	IEC 60814	To be based on LCSET refer Table 1 (i)	12
		IEC 60814	30 mg/kg (for bulk supply), Max	14
vii)	Breakdown voltage	IS : 6792	30 kV (as delivered), Min	40 - 60
		IS : 6792	70 kV (After laboratory treatment), Min	> 72
viii)	Density at 20 °C	IS : 1448 (Part 16)	0.895 g/ml, Max	0.861
	Density at 29.5 °C	IS : 1448 (Part 16)	-	0.856
ix)	DDF at 90 °C	IEC 60247 OR IS : 16086	0.005, Max	0.0004
x)	Particle content	IS:13236	No general requirement	-
B - Refining / Stability				
xi)	Appearance	-	Clear, free from sediment and suspended matter	Clear, free from sediment & suspended matter
xii)	Acidity	IEC 62021 - 1	0.01 mg KOH/g, Max	0.0014
xiii)	Interfacial Tension	ASTM D 971	No general requirement	48
xiv)	Total Sulphur Content	ISO 14596 or ASTM D 4294	No general requirement	-
xv)	Corrosive Sulphur	DIN 51353	Not corrosive	Non corrosive
xvi)	Potentially Corrosive Sulphur	IS: 16310	Not corrosive	Non corrosive
xvii)	DBDS	IS: 16497 (Part 1)	Not detectable (< 5 mg/kg)	Not detectable
xviii)	Inhibitors according to IS : 13631/ IEC 60666	IS: 13631	(U) Uninhibited Oil ; Not detectable (< 0.01%)	Not detectable
xix)	Metal passivator additives according to IS 13631 / IEC 60666	IS: 13631	Not detectable (< 5 mg/kg) or as agreed upon with the purchaser	Not detectable
xx)	Other additives	-	See 7	Not added
xxi)	2 - Furfural & related comp.content	IS : 15668	Not detectable (< 0.05mg/kg) for each individual compound	Not detectable
C - Performance :				
xxii)	Oxidation Stability	IS 12422(Method C) Test duration : (U) Uninhibited Oil : 164 h		
a)	Total acidity	1.9.4 of IS 12422	1.2 mg KOH/g, Max	0.45
b)	Sludge	1.9.1 of IS 12422	0.8 %, Max	0.30
c)	DDF at 90 °C	1.9.6 of is 12422	0.5, Max	0.24
xxiii)	Gassing tendency	IEC 60628, Method A	No general requirement	-
xxiv)	ECT	See 6.14	No general requirement	-
D - Health , safety and environment (HSE)				
xxv)	Flash point	IS 1448 (Part 21)	135 °C, Min	150
xxvi)	PCA content	IP 346	3.0 %, Max	0.28
xxvii)	PCB content	IS 16082	Not detectable (< 2 mg/kg)	Not detectable


Authorised Signatory